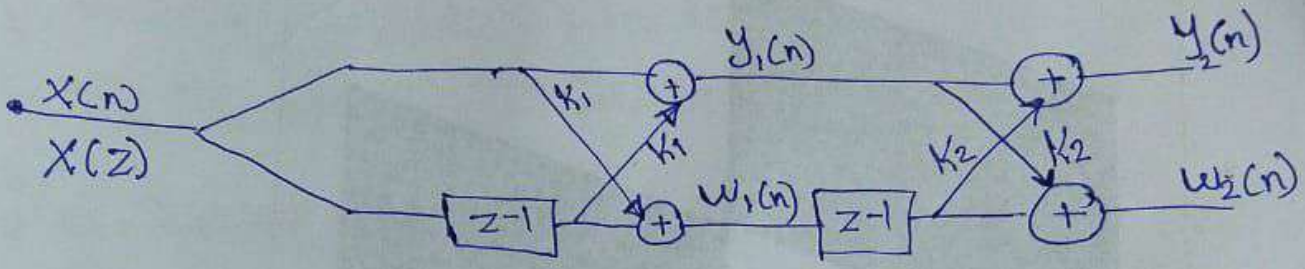
Chapter #5

5.1 The coefficients of a digital Lattice Filter are

$K_1 = -0.266$ and $K_2 = 0.69$

(a) Draw the realization diagram for Lattice structure filter



(b) Calculate the impulse response coefficient for the filter

- The output of stage #1

$$y_1(n) = x(n) + K_1 x(n-1) \longrightarrow (1)$$

$$w_1(n) = K_1 x(n) + x(n-1) \longrightarrow (2)$$

- The output of stage #2

$$y_2(n) = y_1(n) + K_2 w_1(n-1) \longrightarrow (3)$$

For eq (2)

$$w_1(n) = K_1 x(n) + x(n-1) \Longrightarrow w_1(n-1) = K_1 x(n-1) + x(n-2)$$

at eq (3)

$$y_2(n) = x(n) + K_1 x(n-1) + K_2 [K_1 x(n-1) + x(n-2)]$$

$$= x(n) + K_1 x(n-1) + K_1 K_2 x(n-1) + K_2 x(n-2)$$

$$y_2(n) = x(n) + (K_1 + K_1 K_2) x(n-1) + K_2 x(n-2) \longrightarrow (4)$$

(c) Draw its equivalent Transversal FIR Filter structure.

- The Form of the FIR Filter

$$H(z) = h(0) + h(1)z^{-1} + h(2)z^{-2}$$

$$\frac{Y(z)}{X(z)} = h(0) + h(1)z^{-1} + h(2)z^{-2}$$

$$Y(z) = h(0)X(z) + h(1)X(z)z^{-1} + h(2)X(z)z^{-2}$$

↳ Take the inverse Z-Transform

$$y(n) = h(0)x(n) + h(1)x(n-1) + h(2)x(n-2) \longrightarrow (5)$$

by Comparing eq(4) & eq(5)

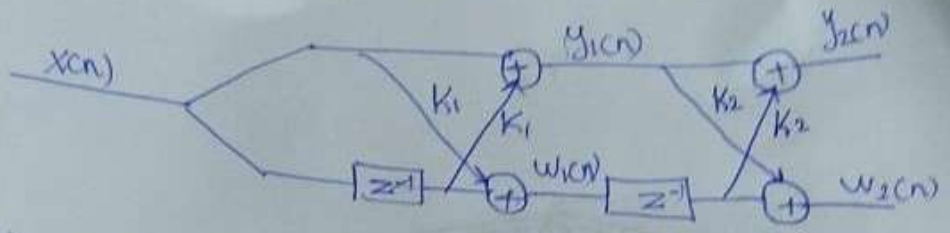
$h(0) = 1$, $h(1) = K_1K_2 + K_1$ $h(2) = K_2$

$h(1) = K_1[1 + K_2]$, $h(2) = K_2$

$= -0.266[1 + 0.69]$ $h(2) = 0.69$

$= -0.449$

(d) Draw its equivalent Lattice structure



$y_1(n) = x(n) + K_1 x(n-1)$

$w_1(n) = K_1 x(n) + x(n-1)$

$y_2(n) = y_1(n) + K_2 w_1(n-1)$

$y_2(n) = x(n) + K_1 x(n-1) + K_2[K_1 x(n-1) + x(n-2)]$

$y_2(n) = x(n) + K_1 x(n-1) + K_1 K_2 x(n-1) + K_2 x(n-2)$

$= x(n) + (K_1 K_2 + K_1) x(n-1) + K_2 x(n-2)$

$\Rightarrow y(n) = h_0 x(n) + h_1 x(n-1) + h_2 x(n-2)$

$K_2 = h_2 = 0.69$

$K_1 K_2 + K_1 = h_1$

$K_1[1 + K_2] = h_1$

$$K_1 = \frac{h_1}{1 + K_2} = \frac{h_1}{1 + h_2} = \frac{-0.266}{1 + 0.69}$$

$$= \frac{-0.266}{1.69} = -0.157$$

5.2

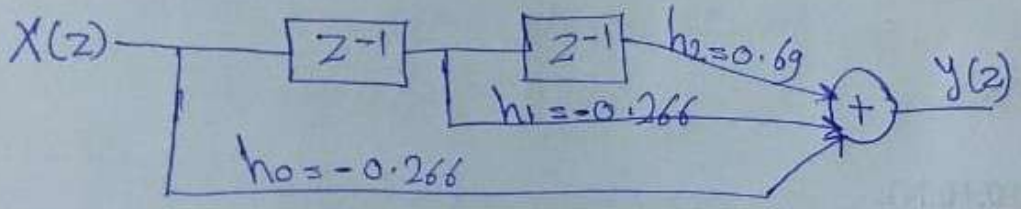
$h_0 = h_1 = -0.266$, $h_2 = 0.69$

(a) Draw the realization structure for this the Transversal Filter.

$$H(z) = h_0 + h_1 z^{-1} + h_2 z^{-2}$$

$$\frac{Y(z)}{X(z)} = h_0 + h_1 z^{-1} + h_2 z^{-2}$$

$$Y(z) = h_0 X(z) + h_1 X(z) z^{-1} + h(2) X(z) z^{-2}$$



(b) Find the Difference eQuation

$$Y(z) = h_0 X(z) + h_1 X(z) z^{-1} + h_2 X(z) z^{-2}$$

Take izt

$$y(n) = h_0 x(n) + h_1 x(n-1) + h_2 x(n-2)$$

(c) Find the Transferfunction, H(z)

$$H(z) = h_0 + h_1 z^{-1} + h_2 z^{-2} = -0.266 + 0.266 z^{-1} + 0.69 z^{-2}$$